Stein's Theorem in General Relativity:
locality, not symmetry, fixes what has become

Working manuscript

25 October 2026

Abstract

Stein proved that in Minkowski spacetime a transitive relation of becoming that is invariant under the spacetime's symmetries, and that makes some past event definite for an event, must be the causal past or degenerate. Clifton and Hogarth showed that, assuming only that the chronological past has become, the options are the causal past, the chronological past and the universal relation; they extended the result to worldline-dependent becoming and remarked that there is "no obvious way" to extend it beyond Minkowski spacetime. We give such an extension and locate exactly what it depends on. In a single curved spacetime, symmetry-invariance is too weak: generic spacetimes have no symmetries, and transitive relations containing the chronological past can then relate spacelike events. The right replacement is local covariance in the sense of locally covariant quantum field theory: one rule of becoming for all globally hyperbolic spacetimes, compatible with isometric embeddings of causally convex regions. Under that hypothesis we prove, by a gluing argument, that either everything has become for everything in every spacetime, or in every spacetime becoming lies between the chronological and the causal past; the worldline-dependent form follows. The residual freedom is exactly the status of lightlike-separated pairs, and we classify it: it can depend on curvature along the connecting light ray, so uniqueness fails even for relations determined by conformal structure alone; topological regularity removes it. In causal set theory the same hypothesis yields an exact Stein theorem, although the analogue fails, as Wüthrich and Callender showed, under invariance within a single causal set. Locality also isolates Gödel-type universes: no morphism connects them to globally hyperbolic spacetimes, so universal becoming there constrains nothing in those. Finally we show why the result does not transfer routinely to spacetimes restricted by energy conditions, and isolate the open problem on which such a transfer depends.

1 Introduction

Stein's theorem is the strongest positive result in the debate over relativistic becoming (Stein, 1968, 1991). Read as a constraint, it says that an objective, transitive becoming relation adapted to Minkowski geometry can only be past causal connectibility, apart from trivial or universal relations. Clifton and Hogarth gave the result its now-standard form (Clifton and Hogarth, 1995): their Theorem 1 assumes invariance under the time-orientation-preserving isometries of Minkowski spacetime, transitivity, and *chronological becoming* (every event in an event's chronological past has become for it), and concludes that becoming is I^-, J^- or the universal relation $\mathcal{U}$. Their Theorem 2 extends this to becoming relative to an observer's worldline. They end with a problem:

> each general relativistic spacetime has its own metric-preserving automorphisms and each its own distinctive geometry, leaving no obvious way to extend Theorem 1 beyond Minkowski spacetime, much less to extend Theorem 2 (Clifton and Hogarth, 1995, p. 385).

They call the question of becoming "in the real world" "more difficult, though more interesting" (ibid., p. 386).

The difficulty is real. Stein's argument runs entirely on symmetry: once one spacelike pair is related, boosts and rotations relate them all, and transitivity collapses the relation to $\mathcal{U}$. A generic curved spacetime has no isometries at all, so the corresponding single-spacetime statement is simply false (Proposition 4). Wüthrich and Callender observed the analogous failure in causal set theory, where invariance under the automorphisms of one causal set permits exotic becoming relations (Wüthrich and Callender, 2017).

Our diagnosis is that the analogue of Minkowskian symmetry in general relativity is not invariance within a spacetime but *uniformity and locality across spacetimes*: one rule, applied in every globally hyperbolic spacetime, whose verdict on a pair of events depends only on a causally convex region containing them. This is the hypothesis of local covariance used to say when a quantum field theory is "the same in all spacetimes" (Brunetti et al., 2003; Fewster and Verch, 2012), and it is closely related to the locality intuition that Viglione defends informally against cosmic simultaneity (Viglione, 2026). With it, Stein's theorem has a curved-spacetime form:

- **No becoming outside the causal past** (Theorem 7). Either becoming is universal in every spacetime, or in every spacetime it lies between I^- and J^-. The worldline-dependent version follows (Corollary 9).
- **The residual freedom is the light cone** (Theorem 10, Theorem 11, Proposition 12). What remains open is exactly which lightlike-separated pairs count. We classify the possibilities; they can depend on Weyl curvature along the connecting light ray, so uniqueness fails even for relations fixed by conformal structure. Closedness or openness of the relation selects J^- or I^-.
- **Locality is indispensable** (Propositions 4 and 5). Natural but non-local rules, such as ordering by a canonical cosmological time, satisfy every other condition and make spacelike events become.
- **Causal sets** (Proposition 19). Under the same hypothesis Stein's theorem holds exactly: becoming is the causal order or universal. Discreteness removes the light-cone freedom.
- **Gödel universes are isolated** (Proposition 20). Totally vicious spacetimes have no proper causally convex regions and embed in no causal spacetime, so local covariance transmits nothing from them.
- **Energy conditions** (Section 8). The proof uses arbitrary globally hyperbolic metrics. Under the dominant energy condition the specific auxiliary spacetime of the proof is provably unavailable, and the transfer reduces to an open gluing problem that we state.

The philosophical upshot is stated in Section 9. What Stein's theorem tells us about relativity survives in general relativity in a robust and a fragile part. The robust part is that nothing outside an event's causal past can have become for it unless everything has. The fragile part is the uniqueness of the answer, which was an artefact of Minkowskian symmetry.

2 Framework

Spacetimes are smooth, connected, four-dimensional, oriented and time-oriented Lorentzian manifolds. We follow Clifton and Hogarth in taking I^-, J^- to be reflexive (Clifton and Hogarth,

1995, p. 360), write $x \ll y$ for $x \in I^-(y)$, $x \neq y$, and $x \leq y$ for $x \in J^-(y)$. We write $I^+_{\mathrm{s}}, I^-_{\mathrm{s}}$ for the strict chronological future and past (defined by timelike curves, so $x \in I^-_{\mathrm{s}}(y) \iff x \ll y$ in causal spacetimes); thus $I^- = I^-_{\mathrm{s}} \cup \Delta$. Standard causal facts are cited from Minguzzi and Sánchez (Minguzzi and Sánchez, 2008), abbreviated MS, whose $I^\pm$ are our strict sets.

Definition 1. GH is the category whose objects are globally hyperbolic spacetimes and whose morphisms $\psi : M \to M'$ are smooth isometric embeddings preserving orientation and time orientation, with $\psi(M)$ open and causally convex in M'. This is the category Loc of locally covariant quantum field theory (Brunetti et al., 2003, §2), (Fewster and Verch, 2012, §2.2).

Definition 2. A *becoming theory* assigns to each $M \in \mathsf{GH}$ a relation $B_M \subseteq M \times M$ (read $x\, B_M\, b$ as "x has become for b") such that, for every M:

(LC) for every morphism $\psi : M \to M'$, $x\, B_M\, b \iff \psi(x)\, B_{M'}\, \psi(b)$;

(T) B_M is transitive;

(CB) $I^-_M \subseteq B_M$.

(CB) is Clifton and Hogarth's chronological becoming; they derive it from "worldline becoming" and "worldline transitivity" (Clifton and Hogarth, 1995, Lemma, p. 364), and the derivation carries over verbatim to curved spacetime. (LC) contains invariance under every orientation- and time-orientation-preserving isometry of each M, which on Minkowski spacetime $\mathbb{M}$ is the proper orthochronous Poincaré group. Its further content is *locality*: if $N \subseteq M$ is open, causally convex and contains x and b, then whether x has become for b is settled by N alone.

Remark 3 (The off-shell premise)**.** GH contains every globally hyperbolic metric, not only solutions of Einstein's equations; the category is defined kinematically in (Brunetti et al., 2003; Fewster and Verch, 2012). Theorem 7 uses this: its proof builds auxiliary spacetimes that need not satisfy any field equation with reasonable matter. This is natural for a kinematic notion of becoming, defined (as Stein and Clifton–Hogarth define theirs) from geometry alone, but it is a premise. Section 8 examines what happens without it.

3 Why one spacetime is not enough

Proposition 4. *Let M be globally hyperbolic and $x, b \in M$ spacelike-separated. Then $R = J^- \cup (J^-(x) \times J^+(b))$, where (e, z) in the second set means e has become for z, is reflexive and transitive, contains I^-, relates x to b, and is neither I^-, J^- nor $\mathcal{U}$. If M has no nontrivial isometries, R is invariant under all of them.*

Proof. Write $P = J^-(x) \times J^+(b)$. Compositions $J^- \circ J^-$, $J^- \circ P$ and $P \circ J^-$ land in J^- or P because $\leq$ is transitive. A composition $P \circ P$ would need some q with $b \leq q \leq x$, which is impossible. The pair (b, x) is not in R, so $R \neq \mathcal{U}$. □

So for a single generic spacetime, Stein-type uniqueness fails completely. The next example shows that uniformity across spacetimes is still not enough without locality.

Proposition 5. *Let $\tau_M(p)$ be the supremum of the lengths of past-directed causal curves from p (the cosmological time of Andersson, Galloway and Howard (Andersson et al., 1998)). Put $B_M = \{(x, b) : \tau_M(x) \leq \tau_M(b)\}$ if τ_M is finite on M, and $B_M = J^-_M$ otherwise. Then B is invariant under all isometries and under the morphisms of GH that are onto, each B_M is reflexive*

and transitive and satisfies (CB), and B relates spacelike pairs in every spacetime where τ is finite (e.g. big-bang Friedmann–Lemaître–Robertson–Walker models). It is not a becoming theory: it violates (LC).

Proof. Isometries preserve lengths of causal curves. If $x \ll b$, concatenating past curves from x with a timelike curve from x to b gives $\tau(b) \geq \tau(x) + d(x,b) > \tau(x)$, so (CB) holds; transitivity is that of $\leq$ on the reals. Failure of (LC): the Milne wedge $W = I^+_{\mathbb{M}}(0)$ is open, causally convex and globally hyperbolic, and its inclusion into $\mathbb{M}$ is a morphism. On W, $\tau_W(p)$ is the proper time from the origin, finite, so B_W relates distinct points of the same hyperboloid $\{\tau_W = c\}$, which are spacelike-separated; on $\mathbb{M}$, $\tau = \infty$ and $B_{\mathbb{M}} = J^-$ does not relate them. □

4 Results

Theorem 6 (Clifton–Hogarth)**.** *For every becoming theory, $B_{\mathbb{M}} \in \{I^-, J^-, \mathcal{U}\}$.*

Proof. Proper orthochronous Poincaré maps are automorphisms of $\mathbb{M}$ in GH, so $B_{\mathbb{M}}$ is invariant under them by (LC). Clifton and Hogarth's Theorem 1 (Clifton and Hogarth, 1995, p. 365) gives the conclusion from this invariance, transitivity and chronological becoming. Its proof (pp. 365–367) uses only point-preserving spatial rotations, boosts and translations, and they note that it does not use spatial reflections (pp. 385–386). Their stated hypothesis is invariance under all time-orientation-preserving isometries, reflections included; using only the proper orthochronous subgroup is a weakening whose adequacy rests on their proof, not their statement. □

Theorem 7 (No becoming outside the causal past)**.** *Let B be a becoming theory.*

(a) If $B_{\mathbb{M}} \neq \mathcal{U}$, then $B_M \subseteq J^-_M$ for every $M \in \mathsf{GH}$.

(b) If $B_{\mathbb{M}} = \mathcal{U}$, then $B_M = \mathcal{U}_M$ for every $M \in \mathsf{GH}$.

Hence either B is universal everywhere, or $I^-_M \subseteq B_M \subseteq J^-_M$ for every M.

Call B *universal* if $B_M = \mathcal{U}_M$ for every M. By Theorem 7 this holds if and only if $B_{\mathbb{M}} = \mathcal{U}$, and if $B_M = \mathcal{U}_M$ for one M then it holds for all; otherwise B is *non-universal*.

Corollary 8 (No shared present)**.** *Call x and b* co-present *in M if $x\, B_M\, b$ and $b\, B_M\, x$. For every non-universal becoming theory, co-presence is the identity relation in every $M \in \mathsf{GH}$. Equivalently, no becoming theory is local, non-trivial and admits two distinct co-present events in some globally hyperbolic spacetime.*

Proof. By Theorem 7(a), $x \leq b$ and $b \leq x$, and globally hyperbolic spacetimes are causal, so $x = b$. □

Corollary 9 (Worldline-dependent becoming)**.** *Suppose that for each $M \in \mathsf{GH}$, each inextendible timelike curve λ in M and each $b \in \lambda$ a set $B_{M,\lambda}(b)$ is given, satisfying Clifton–Hogarth's worldline becoming ($I^-(b) \cap \lambda \subseteq B_{M,\lambda}(b)$) and worldline transitivity ($a \in B_{M,\lambda}(b)$ and $b \in B_{M,\lambda'}(c)$ imply $a \in B_{M,\lambda'}(c)$) (Clifton and Hogarth, 1995, pp. 362–363), and local covariance in the form: for a morphism $\psi : N \to M$, an inextendible timelike λ_N in N and any inextendible timelike λ in M extending $\psi \circ \lambda_N$, $a \in B_{N,\lambda_N}(b) \iff \psi(a) \in B_{M,\lambda}(\psi(b))$. Then $B_{M,\lambda}(b)$ does not depend on λ, and the resulting relation is a becoming theory. Hence Theorem 7 applies to worldline-dependent becoming.*

Clifton and Hogarth anticipated that extending their Theorem 2 would be harder still ("much less to extend Theorem 2", p. 385). Under local covariance it is not: as they themselves note (p. 363), worldline transitivity makes observers whose worldlines cross agree, and the remaining work is done by Theorem 7.

Write $E^- = J^- \setminus I^-$ for the horismos (lightlike-only relation).

Theorem 10 (Classification of the residual freedom)**.** *Let B be a non-universal becoming theory and $N_M = B_M \cap E^-_M$. Then N is locally covariant and satisfies*

(NC) if $(x,y),(y,b) \in N_M$ and $(x,b) \in E^-_M$, then $(x,b) \in N_M$.

Conversely, if $N \subseteq E^-$ is locally covariant and satisfies (NC), then $B = I^- \cup N$ is a becoming theory. On $\mathbb{M}$, N is empty or all of E^-.

Theorem 11 (Non-uniqueness)**.** *Let $N_M(x,b)$ hold iff $x \in E^-(b)$, $x \neq b$, and some lightlike geodesic segment from x to b contains a point at which the Weyl tensor is nonzero. Then $I^- \cup N$ is a becoming theory equal to I^- on $\mathbb{M}$, equal to J^- on any spacetime with nowhere-vanishing Weyl tensor (such as the Schwarzschild exterior), and strictly between I^- and J^- on spacetimes that are flat on a nonempty causally convex open set and have nonzero Weyl tensor somewhere. It depends only on the conformal structure (not merely the curvature: conformally flat spacetimes such as FLRW give I^-).*

Proposition 12 (Regularity)**.** *Let B be non-universal. If every B_M is closed in $M \times M$, then $B = J^-$. If every $B_M \setminus \Delta_M$ is open, then $B = I^-$.*

5 Proofs

5.1 Theorem 10, Theorem 11, Proposition 12

We use: I^+_s is open in $M \times M$ (MS Prop. 2.16); $\overline{J^+} = \overline{I^+_s}$ and $\operatorname{Int} J^+ = I^+_s$ (MS Prop. 2.17), and likewise for the pasts; if $p \ll r \leq q$ or $p \leq r \ll q$ then $p \ll q$, and if $r \in E^+(p)$, $q \in E^+(r)$ then $q \in E^+(p)$ or $p \ll q$ (MS Prop. 2.31); two events joined by a causal curve that is not a maximizing lightlike curve are joined by a timelike curve, and a maximizing lightlike curve is a lightlike geodesic (MS Thm. 2.30(i), (ii)); so a causal curve between horismotically related events is a lightlike geodesic.

Proof of Theorem 10. By Theorem 7(a), $B = I^- \cup N$. Timelike and causal curves between points of a causally convex set stay in it, so I^-, J^- and E^- are locally covariant and N inherits (LC); (NC) is transitivity restricted to horismotic pairs. Conversely, $I^- \cup N$ is reflexive and satisfies (CB); for transitivity, a composition with a chronological step is chronological by MS 2.31, and a composition of two N-steps is chronological or horismotic, in which case (NC) applies. The last claim is Theorem 6. □

Proof of Theorem 11. (NC): if $x \to y \to b$ along lightlike segments γ_1, γ_2 and $x \in E^-(b)$, then $\gamma_1\gamma_2$ is a causal curve between horismotic points, hence an unbroken lightlike pregeodesic (MS Thm. 2.30), and it contains the Weyl point of γ_1. (LC): lightlike geodesics between points of a causally convex image stay in it, and isometries preserve the Weyl tensor. On $\mathbb{M}$ the Weyl tensor vanishes. If it vanishes nowhere, every horismotic pair is in N. If M is flat on an open causally convex V and the Weyl tensor is nonzero at some point z, horismotic pairs inside V are not in N (every causal curve between them stays in V), while a horismotic pair joined by a lightlike segment

through a neighbourhood of z is. The Weyl tensor, as a $(1,3)$ tensor, is invariant under conformal rescaling (standard), and unparametrized lightlike pregeodesics are conformally invariant (MS §2.6). □

Proof of Proposition 12. If B_M is closed and contains I^-, it contains $\overline{I^-} = \overline{I^-_s} = \overline{J^-} \supseteq J^-$, so $B = J^-$ by Theorem 7(a). If $B_M \setminus \Delta$ is open and contained in J^-, it lies in $\operatorname{Int} J^- = I^-_s$, so $B_M \subseteq I^-_s \cup \Delta = I^-$, and (CB) gives $B = I^-$. □

Proof of Corollary 9. By worldline becoming, $b \in B_{M,\lambda'}(b)$ for every $\lambda' \ni b$; with $c = b$, worldline transitivity gives $B_{M,\lambda}(b) \subseteq B_{M,\lambda'}(b)$, and by symmetry they are equal. Call the common set $B_M(b)$. Transitivity: if $a \in B_M(b)$ and $b \in B_M(c)$, choose $\lambda \ni b$, $\lambda' \ni c$ and apply worldline transitivity. (CB): if $a \ll b$, a timelike curve from a to b extends to an inextendible timelike curve λ, and $a \in I^-(b) \cap \lambda \subseteq B_M(b)$. (LC): for $\psi : N \to M$ take $\lambda_N \ni b$ and an inextendible extension λ of $\psi \circ \lambda_N$; then $a \in B_N(b) \iff a \in B_{N,\lambda_N}(b) \iff \psi(a) \in B_{M,\lambda}(\psi(b)) \iff \psi(a) \in B_M(\psi(b))$. □

5.2 Theorem 7

The idea: a forbidden pair in a curved spacetime is moved, by local covariance, into an auxiliary spacetime that agrees with the original near the pair and is flat elsewhere; chronological becoming and transitivity then produce a forbidden pair in a flat region, contradicting Theorem 6.

Lemma 13 (Future pairs reduce to spacelike pairs)**.** *If $x \in J^+(b) \setminus \{b\}$, there is $e \ll x$ spacelike to b.*

Proof. Let U be a convex normal neighbourhood of b and $V \subseteq U$ an open neighbourhood of b that is causally convex in M (M is strongly causal; MS Remark 2.12 and the construction following it). Causal relations in U, regarded as a spacetime, are read off $\exp^{-1}$ (MS Prop. 2.10), and for points of V they agree with those of M (MS Remark 2.12). A future causal curve from b to x is nonconstant (M is causal), so it contains some $b_1 \in V \setminus \{b\}$; then $b \le b_1 \le x$ and $u = \exp_b^{-1}(b_1)$ is a nonzero future causal vector. Choose a spacelike $a \in T_bM$ with $g(u,a) > 0$ and put $q_\varepsilon = \exp_b(\varepsilon a)$. Synge's world function σ is smooth on $U \times U$ with $\nabla_q \sigma(q, b_1)|_{q=b} = -u$, so $2\sigma(q_\varepsilon, b_1) = g(u,u) - 2\varepsilon g(u,a) + O(\varepsilon^2) < 0$ for small $\varepsilon > 0$. The vector $\exp_{q_\varepsilon}^{-1}(b_1)$ depends continuously on ε and tends to $u \neq 0$, so for small ε it is future-directed timelike: $q_\varepsilon \ll b_1$. Also $2\sigma(q_\varepsilon, b) = \varepsilon^2 g(a,a) > 0$, so q_ε is spacelike to b. By MS Prop. 2.31(i), $e = q_\varepsilon \ll x$. □

The same local argument shows that every point has spacelike-separated points arbitrarily close to it.

Lemma 14 (Spacelike chains)**.** *In connected globally hyperbolic M, any two points are joined by a finite chain of consecutively spacelike-separated points.*

Proof. Globally hyperbolic spacetimes (MS Def. 3.70) are causally simple (MS Prop. 3.71), so $J^+(z)$ and $J^-(z)$ are closed and the set of points spacelike to z is open. Fix p and let S be the set of points reachable from p. S is open: if $q \in S$, pick z spacelike to q; a neighbourhood of q is spacelike to z and reachable via q, z. $M \setminus S$ is open: if $q \notin S$ and c is spacelike to q, a neighbourhood W of q is spacelike to c; were some $q'' \in W$ in S, then $c \in S$ and $q \in S$. M is connected. □

The example $M = \mathbb{R} \times S^3$, $b = (0,p)$, $x = (3\pi, p)$ shows that a single point spacelike to two causally related points need not exist; the chain is needed.

Set-up. Fix spacelike-separated $x, b \in M$. Since $J^+ \subseteq M \times M$ is closed (MS Prop. 3.68(iii), 3.71), there are small closed spacelike embedded 3-discs $D_x \ni x$, $D_b \ni b$, each acausal, with no causal curve between them; so $D_x \cup D_b$ is a compact acausal spacelike submanifold with boundary, and Bernal and Sánchez give a smooth spacelike Cauchy hypersurface $S \supseteq D_x \cup D_b$ (Bernal and Sánchez, 2006, Thm. 1.1, Rem. 4.14). Choose $\delta_0 > 0$ and a smooth embedded curve $c : [-\delta_0, L + \delta_0] \to S$, unit speed for the induced metric h, with $c(0) = x$, $c(L) = b$.

Lemma 15 (Adapted coordinates)**.** *Let n be the future unit normal to S, $(e_1 = c', e_2, e_3)$ an h-orthonormal frame along c with (n, e_1, e_2, e_3) positively oriented, $\sigma(s, y) = \exp^h_{c(s)}(y_2 e_2 + y_3 e_3)$, and $\Phi(t, s, y) = \exp^g_{\sigma(s,y)}(t\, n)$. Write $T_\rho = \{|t| < \rho,\ -\rho < s < L + \rho,\ |y| < \rho\}$. For some $\rho_0 \le \delta_0$, Φ is a diffeomorphism from $\overline{T_{\rho_0}}$ onto a closed neighbourhood of $c([0, L])$, orientation- and time-orientation-preserving; $\Phi(\{t = 0\}) \subseteq S$ while $\Phi(t, \cdot) \in I^+(S)$ for $t > 0$ and $\in I^-(S)$ for $t < 0$; $\Phi^* g = -dt^2 + h_t(s, y)$; and $\|\Phi^* g - \eta\| \le K(|t| + |y|)$ on $\overline{T_{\rho_0}}$.*

Proof. Gaussian normal coordinates to S composed with Riemannian normal coordinates of S along the compact embedded arc; points $\Phi(t, \cdot)$ lie on the normal timelike geodesic through a point of the acausal S. At $(t, y) = (0, 0)$ the metric is exactly η because the frame is orthonormal on c; the estimate is smoothness on a compact set. □

Lemma 16 (Cone control)**.** *Let $\eta_c = -c^2 dt^2 + ds^2 + |dy|^2$, $\frac{1}{2} < \mu < 1 < \lambda$, and let k be a symmetric bilinear form on $\mathbb{R}^4$. If $\|k - \eta_1\| \le \varepsilon < \min\{(1 - \mu^2)/(1 + \mu^2), (\lambda^2 - 1)/(\lambda^2 + 1)\}$ (operator norm), then k is Lorentzian with ∂_t timelike, and for $v = (1, u)$: $|u| \le \mu \Rightarrow k(v, v) < 0$, $|u| \ge \lambda \Rightarrow k(v, v) > 0$. The bound is preserved by $\chi k + (1 - \chi)\eta_1$, $0 \le \chi \le 1$.*

Proof. $k(v, v) \le -1 + |u|^2 + \varepsilon(1 + |u|^2)$ and $k(v, v) \ge (1 - \varepsilon)|u|^2 - (1 + \varepsilon)$. The spatial block is $\ge (1 - \varepsilon) I > 0$ and $k(\partial_t, \partial_t) < 0$, which forces signature $(-, +, +, +)$. □

Constants and the auxiliary spacetime. In this order: fix $\mu, \lambda, \varepsilon$ as in Lemma 16; Lemma 15 gives ρ_0 and K; choose $r > 0$ with $3r < \rho_0$, $6Kr \le \varepsilon$ and $8\lambda r < L$. Then $\overline{T_{2r}} \subset T_{3r}$; let χ be smooth, $\chi = 1$ on $\overline{T_{2r}}$, $\operatorname{supp} \chi \subset T_{3r}$, and on $\mathbb{R}^4$ put $g' = \chi\, \Phi^* g + (1 - \chi)\eta$. On T_{3r}, $\|\Phi^* g - \eta\| \le 6Kr \le \varepsilon$, so Lemma 16 applies to g' everywhere and to $\Phi^* g$ on T_{3r}. Every g'-causal vector therefore has $|d(s, y)/dt| \le \lambda$; t increases strictly along future causal curves; and an inextendible causal curve along which t had a finite limit would have convergent spatial coordinates and extend. Hence every level $\{t = c\}$ is a Cauchy hypersurface and $M' = (\mathbb{R}^4, g')$ is globally hyperbolic (MS Thm. 3.75), flat outside T_{3r}, and $\psi = \Phi^{-1} : \Phi(T_{2r}) \to T_{2r}$ is an isometry preserving orientation and time orientation.

Lemma 17 (The slab region is self-determined)**.** *Let $B_0 = \{s \in [-r/2, L + r/2],\ |y| \le r/2\}$, $\Sigma = \Phi(\{0\} \times B_0) \subset S$ and $\Sigma' = \{0\} \times B_0$. Then:*

(i) $D_M(\Sigma) \subseteq \Phi(T_{2r})$ and $D_{M'}(\Sigma') \subseteq T_{2r}$;

(ii) $\psi(D_M(\Sigma)) = D_{M'}(\Sigma')$;

(iii) $N = \operatorname{int} D_M(\Sigma)$ is causally convex in M, globally hyperbolic, connected, and contains x, b; likewise $\psi(N) = \operatorname{int} D_{M'}(\Sigma')$ in M'.

Proof. Escape paths. Let $q \in \partial T_{2r}$ with $0 < t(q) \le 2r$. If the projection (s_q, y_q) is not in B_0, follow the vertical segment down to $t = 0$. Otherwise q lies on the lid $t = 2r$ with $|y_q| \le r/2$; move with velocity $(dt, dy) = (-1, \frac{\mu}{2}\hat{y})$ for coordinate time $2r$, where $\hat{y}$ is a unit vector $(y_q/|y_q|$ if

$y_q \neq 0$), ending at $|y| \in (r/2, 3r)$ because $\mu r > r/2$ and $r/2 + \mu r < 3r$. Either path lies in T_{3r}, has coordinate speed $< \mu$, so is timelike for $\Phi^* g$ and for g' (Lemma 16), and ends at a point of $\{t = 0\}$ outside Σ', i.e. on $S \setminus \Sigma$ in M (Φ is injective). A past-directed causal curve starting on a Cauchy hypersurface never returns to it. The time-reversed paths serve points with $t(q) < 0$.

(i) Let $p \in D^+_M(\Sigma)$ with $p \notin \Phi(T_{2r})$, and let γ be a past-inextendible causal curve from p. It meets Σ; let σ_1 be its first point on Σ, and q the last point of $\partial\Phi(T_{2r})$ on the segment from p to σ_1. After q the segment lies in $\Phi(T_{2r})$, where t increases along future causal curves, so $t(q) > 0$. The segment $p \to q$, the escape path from q, and any past-inextendible continuation form a past-inextendible causal curve from p missing Σ, a contradiction. Time-reverse for D^-. The same argument works in M', where $\{t = 0\}$ is a Cauchy hypersurface.

(ii) Let $p \in D^+_M(\Sigma)$, so $p \in \Phi(T_{2r})$ by (i), and let γ' be a past-inextendible g'-causal curve from $\psi(p)$. If $\psi(p) \in \Sigma'$ there is nothing to prove; otherwise $t(\psi(p)) > 0$. Along γ', $t \to -\infty$; let τ_0 be the first parameter at which $t = 0$ and τ_{exit} the first at which γ' meets ∂T_{2r}. The chart and the agreement $g' = \Phi^* g$ extend to a neighbourhood of $\overline{T_{2r}}$, so segments of γ' up to either time pull back by Φ. If $\tau_0 \leq \tau_{\text{exit}}$ (equality included): were $\gamma'(\tau_0) \notin \Sigma'$, the pulled-back segment ends on $S \setminus \Sigma$ and, continued past S, is a past-inextendible M-curve from p missing Σ (points with $t > 0$ are not on S). If $\tau_{\text{exit}} < \tau_0$, then $t > 0$ at the exit point, and the pulled-back segment followed by the escape path in M again misses Σ. Both contradict $p \in D^+_M(\Sigma)$, so γ' meets Σ'. The reverse inclusion is the same argument with M and M' exchanged, using $D_{M'}(\Sigma') \subseteq T_{2r}$: no global coordinate t on M is needed, since a past-inextendible M-curve from a point of $\Phi(T_{2r})$ either leaves $\Phi(T_{2r})$ or reaches the Cauchy surface S, and the same two cases apply. D^- is time-reversed.

(iii) For $q \in D^+(\Sigma)$, $J^-(q) \cap J^+(\Sigma) \subseteq D^+(\Sigma)$: a past-inextendible curve from r in this set, prefixed by a causal curve $q \to r$, meets Σ; if only on the prefix, at z, then $\sigma \leq r \leq z$ for some $\sigma \in \Sigma$, and acausality gives $r = z \in \Sigma$. Hence $D(\Sigma)$ is causally convex: if $p, q \in D(\Sigma)$, r lies on a causal curve $p \to q$, and an inextendible causal curve λ through r misses Σ, then in the mixed case $p \in D^-$, $q \in D^+$ the curves $(p \to r) \cdot \lambda^+$ and $\lambda^- \cdot (r \to q)$ meet Σ on $p \to r$ and on $r \to q$, and acausality forces both meeting points to be r; the cases $p, q \in D^\pm$ follow from the inclusion just proved. For interior points choose $p^- \ll p$ and $q^+ \gg q$ in $\operatorname{int} D$; then $J(p, q) \subseteq I^+_{\text{s}}(p^-) \cap I^-_{\text{s}}(q^+) \subseteq D(\Sigma)$, an open set (MS Props. 2.16, 2.31), so N is causally convex, its causal diamonds are the compact diamonds of M, and N is globally hyperbolic (MS Def. 3.70). In M', $D_{\eta_\lambda}(\Sigma') \subseteq D_{g'}(\Sigma') \subseteq D_{\eta_\mu}(\Sigma')$ (narrower cones give larger domains of dependence). The first inclusion puts every relative-interior point of Σ', in particular x and b, in the interior; the second puts the projection of every interior point with $t \neq 0$ into $\operatorname{int} B_0$, so its vertical segment to $\{t = 0\}$ lies in $I^-(q^+) \cap I^+(z^-) \subseteq N'$ for suitable q^+, z^-, joining it inside N' to the connected set $\operatorname{int} \Sigma'$. Transfer by (ii). □

By Lemma 17, N is an object of GH and the inclusion $N \hookrightarrow M$ and $\psi|_N : N \to M'$ are morphisms. By (LC),

$$x \, B_M \, b \iff x \, B_N \, b \iff \psi(x) \, B_{M'} \, \psi(b). \tag{1}$$

Lemma 18 (Flat detour)**.** *Let P, Q lie in the flat region of M', be η_λ-spacelike, and be joined by a smooth embedded arc κ contained in a hyperplane $H = \{t = f(s)\}$ with f affine, $|f'| < 1/\lambda$, and at positive distance from $\overline{T_{3r}}$. Then P, Q are spacelike-separated in M', and there is a compact acausal slab $\Sigma'' \subset H$, containing P, Q in its relative interior, such that $N'' = \operatorname{int} D_{M'}(\Sigma'')$ lies in the flat region, equals $\operatorname{int} D_{\mathbb{M}}(\Sigma'')$ under the identity, and is an object of GH whose inclusions into M' and $\mathbb{M}$ are morphisms. Hence $P \, B_{M'} \, Q \iff P \, B_{\mathbb{M}} \, Q$.*

Proof. A g'-causal curve between P, Q would be η_λ-causal. Along g'-causal curves $\frac{d}{d\tau}(t - f(s)) \geq \dot{t}(1 - \lambda|f'|) > 0$ and $t - f(s) \to \pm\infty$, so H is a Cauchy hypersurface for g' and for η. Extend κ

slightly past P and Q and let Σ'' be its closed δ-neighbourhood in H, with δ small compared with the distance from κ to $\overline{T_{3r}}$ and with the curvature radius of κ in H. A Lorentz boost (an isometry of η) makes H horizontal; there the metric is exactly η on a $C\delta$-neighbourhood of Σ'', and the argument of Lemma 17, with the straight tube replaced by the 2δ-tube about the boosted arc and outward escape along the gradient of the distance to the arc, applies with (M, M') replaced by $(M', \mathbb{M})$. It shows that both domains of dependence lie in that neighbourhood, that they coincide, and that their interiors are connected, causally convex and globally hyperbolic. □

Proof of Theorem 7. (a) Suppose $B_{\mathbb{M}} \neq \mathcal{U}$; by Theorem 6, $B_{\mathbb{M}} \subseteq J^-$ relates no spacelike pair. Let $x\, B_M\, b$ with x, b spacelike-separated, and transfer by (1). Take $w = (-4r, 0, 0, 0)$, $v = (4r, L, 0, 0)$ and the arc $t(s) = -4r + (8r/L)s$, $y(s) = (10r \sin(\pi s/L), 0)$, $0 \le s \le L$: where $|t| \le 3r$, $s/L \in [1/8, 7/8]$ and $|y| \ge 10r \sin(\pi/8) > 3r$, elsewhere $|t| > 3r$, and the slope $8r/L < 1/\lambda$. Vertical segments are η_μ-timelike, so $w \ll' \psi(x)$ and $\psi(b) \ll' v$; (CB) gives $w\, B'\, \psi(x)$ and $\psi(b)\, B'\, v$, and (T) gives $w\, B'\, v$. Lemma 18 gives $w\, B_{\mathbb{M}}\, v$ with w, v spacelike, a contradiction. If $x\, B_M\, b$ with $x \in J^+(b) \setminus \{b\}$, Lemma 13 gives $e \ll x$ spacelike to b, and $e\, B_M\, b$ by (CB) and (T), reducing to the previous case.

(b) Suppose $B_{\mathbb{M}} = \mathcal{U}$. For spacelike $x, b \in M$, transfer to M' and use $x' = (4r, 0, 0, 0)$, $b' = (-4r, L, 0, 0)$ with the arc $t(s) = 4r - (8r/L)s$, same $y(s)$: $\psi(x)\, B'\, x'$ and $b'\, B'\, \psi(b)$ by (CB), and $x'\, B'\, b'$ by Lemma 18, since $B_{\mathbb{M}} = \mathcal{U}$; so $x\, B_M\, b$. By Lemma 14 and (T), every pair is related. □

6 Causal sets: an exact Stein theorem

Let CS have locally finite posets as objects and order-embeddings with order-convex image as morphisms. A becoming theory on CS is defined as before, with (CB) read as $x \le b \Rightarrow x\, B\, b$.

Proposition 19. *Every becoming theory on* CS *is either the causal order in every causal set or the universal relation in every causal set.*

Proof. If x, b are incomparable, $\{x, b\}$ is convex, so both embeddings of the two-element antichain are morphisms: the verdict on incomparable pairs is a single symmetric bit β, the same in every causal set. If $x > b$, the interval $I = [b, x]$ is finite and convex, so $x\, B\, b$ depends only on I. If $\beta = 1$, in $I \sqcup \{z\}$ with z incomparable to all, $x\, B\, z\, B\, b$; so every such verdict is positive and B is universal. If $\beta = 0$ and $x\, B\, b$ for some $x > b$, adjoin e with $e < x$ only: I remains convex, so $x\, B\, b$ there, $e\, B\, x$ by (CB), and $e\, B\, b$ with e incomparable to b, contradicting $\beta = 0$. □

Wüthrich and Callender state that "The analogue of this theorem, however, is straightforwardly false in CST" (Wüthrich and Callender, 2017, §3). Their counterexample, on the diamond causal set $d < b, c < a$, adds the incomparable pair $\{b, c\}$ to the causal order and is invariant under that causal set's automorphisms. It is not the restriction of any becoming theory in the present sense: relating b to c fixes $\beta = 1$, and Proposition 19 then relates b to d, which their relation excludes. Their own footnote 5 observes that the counterexample "could be a small proper part of a much larger causet"; (LC) is exactly the requirement that the relation on such a part agree with the relation in the larger causal set. The diagnosis matches the continuum case: invariance within one structure is too weak, and locality across structures restores Stein's theorem, here with no residual freedom because a causal set has no light-cone boundary. Their claim is true on its own terms; the present result uses a stronger, cross-structure hypothesis. Note also that CS includes posets that are not manifoldlike (antichains, disjoint unions), and the proof uses them: this is the causal-set counterpart of the off-shell premise of Remark 3.

7 Gödel universes are isolated

Proposition 20. *Extend* GH *to all oriented, time-oriented spacetimes with the same morphisms. If M is totally vicious ($p \in I^-_{\mathrm{s}}(p)$ for all p), then every morphism into M is onto, every theory satisfying (CB) has $B_M = \mathcal{U}_M$, and there is no morphism between M and any causal spacetime, in either direction.*

Proof. In a totally vicious spacetime $I^-(p) = I^+(p) = M$ for every p (MS §3.1). Let V be nonempty, open and causally convex, $p \in V$ and $q \in M$. There are timelike curves $p \to q$ and $q \to p$; their concatenation is a causal curve with both endpoints in V, so it lies in V, and $q \in V$. Thus $V = M$ (compare MS Remark 2.12). (CB) gives $B_M \supseteq I^-_M = M \times M$. A morphism $N \to M$ from a causal N would be onto, hence an isometry, which is impossible since M has closed timelike curves; a morphism $M \to N$ would map a closed timelike curve of M to one of N. □

Gödel's universe is totally vicious, so universal becoming there is forced; this is Gödel's first step, in Stein's terms. But local covariance transmits constraints only along morphisms, and by Proposition 20 no morphism connects a totally vicious spacetime with a globally hyperbolic one. Gödel's modal step, from what holds in such a universe to what holds in ours (Gödel, 1949, pp. 561–562), as quoted in (Monton, 2006, §6), requires that becoming be the same feature across nomologically possible universes. Local covariance is one precise way of requiring a rule to be the same across universes, and it transmits nothing out of totally vicious ones. A version of Gödel's step that does go through would need a different cross-universe link. Proposition 5 shows what such links look like: natural but non-local. Compare the conceptual responses of Savitt (2005), Dieks (2006) and Aames (2022), which locate becoming locally along worldlines.

8 Energy conditions

Let $\mathsf{GH}_{\mathrm{DEC}}$ be the full subcategory of spacetimes whose Einstein tensor satisfies the dominant energy condition. It is closed under restriction to causally convex open subsets and contains $\mathbb{M}$.

Proposition 21 (The auxiliary spacetime is unavailable)**.** *If $M \in \mathsf{GH}_{\mathrm{DEC}}$ is flat on a neighbourhood of some totally geodesic Cauchy surface, then M is flat. In particular M' above lies in $\mathsf{GH}_{\mathrm{DEC}}$ only if g is flat on T_{2r}.*

Proof. On the Cauchy surface S the metric is flat and the Einstein tensor vanishes. By the conservation theorem for matter obeying the dominant energy condition (Hawking and Ellis, 1973, §4.3), the Einstein tensor vanishes on its domain of dependence, i.e. everywhere. In the form we use: let t be a Cauchy temporal function with $S = t^{-1}(0)$ (Bernal and Sánchez, 2006, Thm. 1.2) and $p \in J^+(S)$. The set $J^-(p) \cap J^+(S)$ is compact, and it is the union of the segments of integral curves of ∇t that run from S to $\partial J^-(p)$ (a future-directed curve that leaves the past set $J^-(p)$ does not return). Minguzzi's form of the theorem (Minguzzi, 2015, Thm. 1), which needs no time function, then gives $G = 0$ at p from $G = 0$ on S; $J^-(S)$ is the time reverse. For M', any slice $\{t = c\}$ with $c < -3r$ is totally geodesic and has a flat neighbourhood. A vacuum globally hyperbolic spacetime with flat data $(\Sigma, h, 0)$ on a Cauchy surface is a development of those data and so embeds isometrically in their maximal development (Choquet-Bruhat and Geroch, 1969, Thm. 3). That maximal development is the flat static spacetime $\mathbb{R} \times \Sigma$, $-dt^2 + h$ (not Minkowski in general; e.g. $\Sigma = T^3$): it is a development, and if a development properly extended it, a point p outside its image would lie on a timelike geodesic from Σ whose image under the extension

coincides with the corresponding complete geodesic of $\mathbb{R} \times \Sigma$, so p would lie in the image. Hence the spacetime is flat. □

The same reasoning constrains any repair that keeps the original spacetime and inserts small flat pockets near w, v and the detour: in a vacuum region whose causal past on some Cauchy surface is vacuum and unmodified, the conservation theorem forces the modified spacetime to be vacuum there with the same data, so no such pocket can be inserted there. Pockets must be fed by matter brought in from the past and routed around $D(\Sigma)$ afterwards. The proof of Theorem 7 goes through verbatim in $\mathsf{GH}_{\mathrm{DEC}}$ given the following sufficient condition, which we leave open.

> **Pocket problem.** Given $M \in \mathsf{GH}_{\mathrm{DEC}}$ and Σ as above, find $M'' \in \mathsf{GH}_{\mathrm{DEC}}$ containing an isometric copy of a neighbourhood of $D_M(\Sigma)$ whose image is a domain of dependence, and exactly flat causally convex regions containing points $w \ll x$ and $v \gg b$ joined by the flat domain of dependence of a slab.

The constraint is an inequality, not a field equation, so constructions can be kinematic, and explicit models with flat regions chronologically adjacent to curved dominant-energy regions exist (Vaidya spacetimes before an incoming, or after an outgoing, null-dust shell). Thin shells will not suffice in general, since null junctions require matching intrinsic geometry. A solution would give Theorem 7 for spacetimes with physically reasonable matter; a vacuum version would require bounded interior flat regions adjacent to arbitrary curved vacuum regions, which we have not found in the gluing literature (an abstract-level survey: Carlotto–Schoen, Corvino–Schoen, Aretakis–Czimek–Rodnianski, Czimek–Rodnianski).

9 What this shows about becoming

The robust part of Stein's theorem. In general relativity, under locality, nothing outside an event's causal past can have become for it unless everything has. This holds in every globally hyperbolic spacetime, however asymmetric, and does not rely on any foliation. Those who want an extended, spacelike "now" as part of objective becoming face Stein's dilemma in general relativity too, unless they give up locality or enrich the spacetime structure itself, adding a preferred foliation or reference field so that becoming is local relative to the richer structure (in effect Monton's move). The theorem then does not apply, because the category, and with it the notion of an isometric embedding, has changed; the cost is the added structure.

A trilemma. Suppose a shared present is understood as a set of distinct events each of which has become for each other, the sense in which Stein's dilemma is posed. Then Corollary 8 can be read as a trilemma for dynamic ontologies in general relativity: locality, a shared (extended) present, and non-trivial becoming cannot all be had. Defenders of a shared now give up locality or add structure; local becoming theorists get point-like presents, which recalls Stein's remark that the converse of Putnam's premise leads to "a peculiarly extreme (but pluralistic!) form of solipsism" (Stein, 1968, p. 18); and giving up non-triviality is the block. Other notions of a present, for instance a set of events none of which has become for another, are not addressed by the corollary. The corollary says nothing about which horn to take.

The fragile part. Uniqueness does not survive. Minkowskian symmetry forced a choice between I^- and J^-; general relativity leaves open how lightlike-separated events stand, and a rule may let

curvature along the light ray decide. Clifton and Hogarth's debate over causal versus chronological becoming (Clifton and Hogarth, 1995, pp. 364–365) thus acquires a third family of options; the choice between the endpoints is equivalent to a regularity requirement (Proposition 12).

Locality does the work symmetry did. In a single curved spacetime, definability-style invariance constrains almost nothing (Proposition 4); natural global rules such as cosmological time violate only locality (Proposition 5). So the curved-spacetime case for "only the past has become" rests on a locality principle of the kind locally covariant field theory takes for granted and akin to Viglione's Causal Connection Condition (Viglione, 2026) (which is stricter: (LC) allows dependence on spacelike events inside the convex region). Rejecting locality is a coherent option, but it is the option of global, cosmological becoming, with its own costs (Dieks, 2006; Viglione, 2026).

Limits. The theorem concerns relations of becoming, not ontology; it does not decide between presentism, the growing block and eternalism. It assumes that becoming is defined for every globally hyperbolic geometry (Remark 3); Section 8 shows that dropping this premise is not routine. Spacetimes that are not globally hyperbolic are not treated beyond Proposition 20.

10 Prior work

Stein's theorem and its Clifton–Hogarth form are the direct predecessors (Stein, 1968, 1991; Clifton and Hogarth, 1995). Rakić (1997) takes an event-relative, transitive "realized with respect to" relation, argues that under her constraints it is not definable from the causal geometry of Minkowski spacetime, and adds it as a conservative primitive: in our terms, the added-structure option. Point-based presents (Müller, 2006) and the causal-diamond present of Arthur and Savitt (Arthur, 2006, 2019; Savitt, 2009, 2015), with Dorato's critique (Dorato, 2011) and Aames's application to Gödel universes (Aames, 2022), are conceptual proposals without a curved-spacetime classification. In flat spacetime, definability of sets from relativistic structure is governed by automorphism groups (Madarász et al., 2025). A citation sweep of Clifton and Hogarth (1995) and Stein (1991) (Semantic Scholar, 24 October 2026), and full-text reading of the works named here, found no prior extension of Stein's theorem to curved spacetime. This bounds, but does not certify, priority.

Note on method. This manuscript was developed with substantial assistance from large language models (Anthropic Claude and OpenAI Codex), including literature search, proof drafting and an independent cross-model proof check that found and repaired an error in an earlier draft. Quotations were checked against the primary texts; responsibility for errors is the author's.

References

J. Aames, Temporal becoming in a relativistic universe: causal diamonds and Gödel's philosophy of time, *Eur. J. Philos. Sci.* 12 (2022) 44. doi:10.1007/s13194-022-00471-z.

L. Andersson, G. J. Galloway and R. Howard, The cosmological time function, *Class. Quantum Grav.* 15 (1998) 309–322. doi:10.1088/0264-9381/15/2/006.

R. T. W. Arthur, Minkowski spacetime and the dimensions of the present, in D. Dieks (ed.), *The Ontology of Spacetime*, Elsevier, 2006, ch. 7. doi:10.1016/s1871-1774(06)01007-2.

R. T. W. Arthur, Relativity and the present, in *The Reality of Time Flow*, Springer, 2019, ch. 6. doi:10.1007/978-3-030-15948-1_6.

A. N. Bernal and M. Sánchez, Further results on the smoothability of Cauchy hypersurfaces and Cauchy time functions, arXiv:gr-qc/0512095.

R. Brunetti, K. Fredenhagen and R. Verch, The generally covariant locality principle, *Commun. Math. Phys.* 237 (2003) 31–68. doi:10.1007/s00220-003-0815-7.

Y. Choquet-Bruhat and R. Geroch, Global aspects of the Cauchy problem in general relativity, *Commun. Math. Phys.* 14 (1969) 329–335. doi:10.1007/bf01645389.

R. Clifton and M. Hogarth, The definability of objective becoming in Minkowski spacetime, *Synthese* 103 (1995) 355–387. doi:10.1007/BF01089733.

D. Dieks, Becoming, relativity and locality, in D. Dieks (ed.), *The Ontology of Spacetime*, Elsevier, 2006, ch. 8. doi:10.1016/s1871-1774(06)01008-4.

M. Dorato, The Alexandroff present and Minkowski spacetime: why it cannot do what it has been asked to do, in *Explanation, Prediction, and Confirmation*, Springer, 2011. doi:10.1007/978-94-007-1180-8_26.

C. J. Fewster and R. Verch, Dynamical locality and covariance: what makes a physical theory the same in all spacetimes?, arXiv:1106.4785.

K. Gödel, A remark about the relationship between relativity theory and idealistic philosophy, in P. A. Schilpp (ed.), *Albert Einstein: Philosopher-Scientist*, 1949, pp. 557–562.

S. W. Hawking and G. F. R. Ellis, *The Large Scale Structure of Space-Time*, Cambridge University Press, 1973. doi:10.1017/cbo9780511524646.

J. Madarász, M. Stannett and G. Székely, Definable coordinate geometries over fields, part 1: theory, arXiv:2507.10279.

E. Minguzzi and M. Sánchez, The causal hierarchy of spacetimes, in *Recent Developments in Pseudo-Riemannian Geometry*, ESI Lectures in Mathematics and Physics, EMS, 2008, pp. 299–358. doi:10.4171/051-1/9; arXiv:gr-qc/0609119.

E. Minguzzi, The vacuum conservation theorem, *Gen. Relativ. Gravit.* 47 (2015) 32, arXiv:1406.5915.

B. Monton, Presentism and quantum gravity, in D. Dieks (ed.), *The Ontology of Spacetime*, Elsevier, 2006, ch. 14 (read in the author's 2005 preprint).

T. Müller, On the problem of defining the present in special relativity: a challenge for tense logic, in *Time and History*, De Gruyter, 2006. doi:10.1515/9783110333213.441.

N. Rakić, Past, present, future, and special relativity, *Brit. J. Philos. Sci.* 48 (1997) 257–280. doi:10.1093/bjps/48.2.257.

S. Savitt, Time travel and becoming, *The Monist* 88 (2005) 413–422. doi:10.5840/monist200588323.

S. Savitt, The transient nows, in *Quantum Reality, Relativistic Causality, and Closing the Epistemic Circle*, Springer, 2009. doi:10.1007/978-1-4020-9107-0_18.

S. Savitt, I ♡ ◊s, *Stud. Hist. Philos. Mod. Phys.* 50 (2015) 19–24. doi:10.1016/j.shpsb.2015.02.001.

H. Stein, On Einstein–Minkowski space–time, *J. Philos.* 65 (1968) 5–23. doi:10.2307/2024512.

H. Stein, On relativity theory and openness of the future, *Philos. Sci.* 58 (1991) 147–167. doi:10.1086/289609.

F. Viglione, Not-so-absolute cosmic simultaneity, *Erkenntnis* (2026). doi:10.1007/s10670-026-01075-2.

C. Wüthrich and C. Callender, What becomes of a causal set?, *Brit. J. Philos. Sci.* 68 (2017) 907–925. doi:10.1093/bjps/axv040; arXiv:1502.00018.